

ESOPHAGEAL DISORDERS

Anatomy & physiology

- Musculomembranous tube, **24–34 cm**; add 15 cm → incisors as landmark. Upper + lower esophageal sphincters (functional narrowing).
- **Peristalsis: Primary** (swallow-triggered, brainstem, 2 cm/s) · **Secondary** (triggered by distension/retained bolus/reflux) · **Tertiary** (non-propulsive, incoordinate, moves bolus up/down).

GERD — overview

- Common chronic disorder; prevalence **8–20%**. **Abnormal LES opening** → **mucosal damage**. Both genders equal (except pregnancy).
- **Risk associations:** hiatal hernia, scleroderma, asthma, COPD, diabetes, HRT, coeliac disease.

GERD — complications

- Esophagitis, erosions, **strictures**, **Barrett's esophagus**, **esophageal adenocarcinoma**.

GERD — alarm symptoms (always test)

- **Dysphagia, odynophagia, GI bleeding, vomiting, unexplained weight loss, iron deficiency anaemia, early satiety, choking, continuous pain.**

GERD — diagnosis

- **Clinical history** (classic heartburn → assume GERD = uncomplicated).
- **Empiric PPI trial** (response = diagnostic, per ACG).
- **Endoscopy** = technique of choice to identify complications; **biopsy for Barrett's + adenocarcinoma**. Many GERD patients have normal mucosa. Only for alarm symptoms / refractory.
- **Ambulatory pH monitoring:** excessive acid exposure; useful if not responding.
- **Barium:** not routine (low sensitivity); detects achalasia + hiatal hernia.

GERD — treatment

- **Goals:** relieve symptoms, ↓ recurrence, heal tissue injury, prevent complications.
- **Stepwise:** self-care (antacids, OTC H₂-antagonist/PPI) + lifestyle → no response in 2 weeks → **empiric PPI** → no response or alarm → **endoscopy**.
- **Lifestyle:** weight loss, elevate head of bed 6–8 inches, smaller frequent meals, protein-rich (↑LES pressure), **avoid eating 3h before lying down**, avoid trigger foods/alcohol, **stop tobacco**.
- **Anti-reflux surgery** = last resort (refractory, complications, undesirable long-term meds). Endoscopic therapy still trial.

Achalasia

- Greek = "does not relax". **Spasm of cardia + dilatation of lower 2/3 esophagus**.
- **Etiology:** **denervation of esophageal smooth muscle** (cause unknown). **Trypanosoma cruzi** → **Chagas disease (megaesophagus)**.
- **Symptoms:** progressive **dysphagia (more for solids... actually solids+liquids)**, retrosternal pain, regurgitation, nocturnal cough, weight loss.

- **Diagnosis:** **barium swallow** = "**bird's beak**"; **manometry** (LES pressure, fails to relax); endoscopy to exclude other disease (tight LES).
- **Treatment:**
- Pharmacological: **calcium channel blockers**, **isosorbide dinitrate** (short-term; SE hypotension/headache).
- **Balloon dilatation:** success 60–90%, perforation 1–5%.
- **Botulinum toxin** injection into LES: success 90% but **high recurrence (50%)**.
- **Myotomy** (controlled LES incision): thoracoscopy, laparoscopy, or **POEM (peroral endoscopic myotomy)**.

Diffuse esophageal spasm (DES)

- Fragmental degeneration of vagal nerve; **female > male**. **Simultaneous, repetitive, high-pressure, UNcoordinated contractions**.
- Severe intermittent chest pain, dysphagia, food impaction.
- **Diagnosis:** **barium** = "**corkscrew esophagus**" / **pseudodiverticulum**; manometry = simultaneous multi peaked contractions, high amplitude (120 mmHg) or long duration (>2.5s).
- **Treatment:** acid suppression, nitrates, CCB, sedatives, anticholinergic, botulinum toxin. Surgery (long esophagomyotomy) if severe.

Hypercontractile (nutcracker) esophagus

- Excessively strong but **COORDINATED** peristalsis (unlike DES which is uncoordinated). Abnormal excitatory innervation / heightened sensitivity.
- **Severe squeezing retrosternal chest pain (mimics angina)**, dysphagia to solids + liquids.
- **Diagnosis:** **manometry** = **high-amplitude (>180 mmHg) peristalsis + normal LES relaxation**; barium may show corkscrew; endoscopy normal.
- **Treatment:** avoid triggers; CCB (nifedipine), nitrates, peppermint oil, **low-dose antidepressants** (↓visceral pain). Refractory → botulinum toxin, POEM.

Esophageal perforation/rupture

- **Traumatic (more common):** endoscopy/dilation, **stent placement (most common, up to 25%)**, vomiting/straining (Boerhaave), penetrating trauma.
- **Non-traumatic:** neoplasm/ulceration, caustic ingestion.
- **Pathophysiology:** air/saliva/gastric contents → **mediastinitis, pneumomediastinum, empyema** → sepsis, shock, respiratory failure.
- **Presentation:** lower chest/upper abdomen pain (radiates to L shoulder/back), vomiting/haematemesis, dyspnoea, cough on swallowing, fever, **subcutaneous emphysema**, tachycardia, upper abdominal rigidity, pneumo/hydrothorax, sepsis/shock.
- **Imaging:** CXR — **hydrothorax (L>R), pneumomediastinum, subcutaneous emphysema, mediastinal widening**, subdiaphragmatic air.
- **Contrast:** **Gastrografin (water-soluble) preferred** (less sensitive); barium more sensitive (positive in 22% of non-diagnostic Gastrografin). **No contrast in sedated patients** (need intact gag reflex); use CT if sedated.
- **Management:** ICU, NPO, NG suction, broad-spectrum antibiotics, narcotics. **Early surgery (first 24h) reduces mortality**.

Esophageal carcinoma

- **Squamous cell carcinoma:** upper + mid esophagus; risk = **tobacco + alcohol**.
- **Adenocarcinoma:** lower esophagus; risk = **obesity, GERD, Barrett's esophagus**.

- **Diagnosis:** history/exam, **upper GI endoscopy + biopsy**, CT + PET, **EUS** (depth/staging).
- **Treatment:** early = cure (esophagectomy or endoscopic resection); late = palliative. Chemo-radiotherapy pre/post.
- **Endoscopic resection (EMR / ESD)** for very early-stage / precancerous tissue.

High-yield one-liners

- **GERD:** PPI trial = diagnostic; endoscopy for alarm/refractory; complications = Barrett's → adenocarcinoma.
- **Alarm symptoms** (dysphagia, weight loss, bleeding, anaemia) → always endoscope.
- **Achalasia** = bird's beak (barium) + failed LES relaxation (manometry); treat balloon dilatation / myotomy / POEM / botox.
- **DES** = corkscrew, UNcoordinated. Nutcracker = coordinated, high amplitude >180 mmHg.
- **Perforation:** stent = most common cause; Gastrografin first; early surgery <24h; mediastinitis.
- **SCC** = upper/mid, tobacco+alcohol. Adenocarcinoma = lower, GERD/Barrett's/obesity.

ESOPHAGEAL DISORDERS — Clinical Revision Table

Bold = high-yield. Dr Fahmi Kakamad. Covers GERD, motility disorders, perforation, carcinoma.

GERD				
Overview	Prevalence 8–20%; abnormal LES opening → mucosal damage . Assoc: hiatal hernia, scleroderma, asthma, COPD, DM, HRT, coeliac			
Complications	Esophagitis, erosions, strictures , Barrett's esophagus , adenocarcinoma			
Alarm symptoms	Dysphagia , odynophagia , GI bleed , weight loss , iron-def anaemia , early satiety , vomiting , choking → always endoscope			
Diagnosis	Clinical (heartburn) → empiric PPI trial (response = dx) . Endoscopy + biopsy for complications/Barrett's. pH monitoring if refractory. Barium not routine			
Treatment	Self-care (antacid/H2/PPI) + lifestyle → 2wk no response → empiric PPI → alarm/refractory → endoscopy. Lifestyle: weight loss, elevate bed 6–8", small meals, no eating 3h pre-sleep, stop alcohol/tobacco. Anti-reflux surgery = last resort			
MOTILITY DISORDERS				
Disorder	Pathophysiology	Symptoms	Diagnosis	Treatment
Achalasia	Failure of LES to relax + dilated lower 2/3 (denervation; T. cruzi = Chagas)	Progressive dysphagia (solids+liquids), regurgitation, nocturnal cough, weight loss	Barium = bird's beak; manometry = LES fails to relax ; endoscopy excludes other	CCB/nitrate (short); balloon dilatation (60–90%, perforation 1–5%) ; botox (90% but 50% recurrence); myotomy/POEM
Diffuse esophageal spasm	Vagal degeneration; female > male. Simultaneous, UNcoordinated high-pressure contractions	Severe intermittent chest pain, dysphagia, food impaction	Barium = corkscrew/pseudodiverticulum ; manometry simultaneous multipeaked (120 mmHg / >2.5s)	Acid suppression, nitrate, CCB, sedative, botox. Surgery (long esophagomyotomy) if severe
Nutcracker (hypercontractile)	Excessively strong but COORDINATED peristalsis (unlike DES)	Severe squeezing chest pain (mimics angina), dysphagia solids+liquids	Manometry = high amplitude >180 mmHg + normal LES relaxation ; endoscopy normal	Avoid triggers; CCB (nifedipine), nitrate, peppermint oil, low-dose antidepressant ; refractory → botox/POEM
PERFORATION & CARCINOMA				
	Cause / risk	Features / diagnosis / treatment		
Esophageal perforation	Traumatic (common): stent placement = most common (25%), endoscopy/dilation, vomiting (Boerhaave), trauma . Non-traumatic: neoplasm, caustic. → mediastinitis, pneumomediastinum, empyema, sepsis	Lower chest/upper abd pain (→L shoulder/back), vomiting/haematemesis, subcutaneous emphysema , fever, tachycardia, shock. CXR: hydrothorax L>R, pneumomediastinum, mediastinal widening . Gastrografin first (no contrast if sedated). Mgmt: ICU, NPO, NG suction, broad-spectrum Abx, early surgery <24h		
Esophageal carcinoma	SCC = upper/mid, tobacco + alcohol. Adenocarcinoma = lower, obesity/GERD/Barrett's	Dx: endoscopy + biopsy , CT/PET, EUS (depth/staging) . Tx: early = cure (esophagectomy or endoscopic resection EMR/ESD); late = palliative; chemo-radiotherapy pre/post		